

Python Machine Learning By Example

Fourth Edition Unlock machine learning best practices with real-world use cases Yuxi (Hayden) Liu

Chapter 1, *Getting Started with Machine Learning and Python*, will kick off your Python machine learning journey. It starts with what machine learning is, why we need it, and its evolution over the last few decades. It then discusses typical machine learning tasks and explores several essential techniques of working with data and working with models, in a practical and fun way. You will also set up the software and tools needed for examples and projects in the upcoming chapters.

Chapter 2, *Building a Movie Recommendation Engine with Naïve Bayes*, focuses on classification, specifically binary classification and Naïve Bayes. The goal of the chapter is to build a movie recommendation system. You will learn the fundamental concepts of classification, and about Naïve Bayes, a simple yet powerful algorithm. It also demonstrates how to fine-tune a model, which is an important skill for every data science or machine learning practitioner to learn.

Chapter 3, *Predicting Online Ad Click-Through with Tree-Based Algorithms*, introduces and explains in depth tree-based algorithms (including decision trees, random forests, and boosted trees) throughout the course of solving the advertising click-through rate problem. You will explore decision trees from the root to the leaves, and work on implementations of tree models from scratch, using scikit-learn and XGBoost. Feature importance, feature selection, and ensemble will be covered alongside.

Chapter 4, *Predicting Online Ad Click-Through with Logistic Regression*, is a continuation of the ad click through prediction project, with a focus on a very scalable classification model—logistic regression. You will explore how logistic regression works, and how to work with large datasets. The chapter also covers categorical variable encoding, L1 and L2 regularization, feature selection, online learning, and stochastic gradient descent.

Chapter 5, *Predicting Stock Prices with Regression Algorithms*, focuses on several popular regression algorithms, including linear regression, regression tree and regression forest. It will

encourage you to utilize them to tackle a billion (or trillion) dollar problem—stock price prediction. You will practice solving regression problems using scikit-learn and TensorFlow.

Chapter 6, Predicting Stock Prices with Artificial Neural Networks, introduces and explains in depth neural network models. It covers the building blocks of neural networks, and important concepts such as activation functions, feedforward, and backpropagation. You will start by building the simplest neural network and go deeper by adding more layers to it. We will implement neural networks from scratch, use TensorFlow and PyTorch, and train a neural network to predict stock prices.

Chapter 7, Mining the 20 Newsgroups Dataset with Text Analysis Techniques, will start the second step of your learning journey—unsupervised learning. It explores a natural language processing problem—exploring newsgroups data. You will gain hands-on experience in working with text data, especially how to convert words and phrases into machine-readable values and how to clean up words with little meaning. You will also visualize text data using a dimension reduction technique called t-SNE. Finally, you will learn how to represent words with embedding vectors.

Chapter 8, Discovering Underlying Topics in the Newsgroups Dataset with Clustering and Topic Modeling, talks about identifying different groups of observations from data in an unsupervised manner. You will cluster the newsgroups data using the K-means algorithm, and detect topics using non-negative matrix factorization and latent Dirichlet allocation. You will be amused by how many interesting themes you are able to mine from the 20 newsgroups dataset!

Chapter 9, Recognizing Faces with Support Vector Machine, continues the journey of supervised learning and classification. Specifically, it focuses on multiclass classification and support vector machine classifiers. It discusses how the support vector machine algorithm searches for a decision boundary in order to separate data from different classes. You will implement the algorithm with scikit-learn, and apply it to solve various real-life problems including face recognition.

Chapter 10, Machine Learning Best Practices, aims to fully prove your learning and get you ready for real-world projects. It includes 21 best practices to follow throughout the entire machine learning workflow.

11, Categorizing Images of Clothing with Convolutional Neural Networks, is about using Convolutional Neural Networks (CNNs), a very powerful modern machine learning model, to classify images of clothing. It covers the building blocks and architecture of CNNs, and their implementation using PyTorch. After exploring the data of clothing images, you will develop CNN models to categorize the images into ten classes, and utilize data augmentation and transfer learning techniques to boost the classifier.

Chapter 12, Making Predictions with Sequences using Recurrent Neural Networks, starts by defining sequential learning, and exploring how Recurrent Neural Networks (RNNs) are well suited for it. You will learn about various types of RNNs and their common applications. You will implement RNNs with PyTorch, and apply them to solve three interesting sequential learning problems: sentiment analysis on IMDb movie reviews, stock price forecasting, and text auto-generation.

Chapter 13, Advancing Language Understanding and Generation with the Transformer Models, dives into the Transformer neural network, designed for sequential learning. It focuses on crucial parts of the input sequence and captures long-range relationships better than RNNs. You will explore two cutting-edge Transformer models BERT and GPT, and use them for sentiment analysis and text generation, which surpass the performance achieved in the previous chapter.

Chapter 14, Building an Image Search Engine Using CLIP: A Multimodal Approach, explores a multimodal model, CLIP, that merges visual and textual data. This powerful model can understand connections between images and text. You will dive into its architecture and how it learns, then build an image search engine. Finally, you will cap it all off with a zero-shot image classification project, pushing the boundaries of what this model can do.

Chapter 15, Making Decisions in Complex Environments with Reinforcement Learning, is about learning from experience, and interacting with the environment. After exploring the fundamentals of reinforcement learning, you will explore the FrozenLake environment with a

simple dynamic programming algorithm. You will learn about Monte Carlo learning and use it for value approximation and control. You will also develop temporal difference algorithms and use Q-learning to solve the taxi problem.